# Projective Geometry

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## 1 Introduction

Projective Geometry is essential in solving hard geometry problems. While it seems customary to teach inversion along with projective geometry, and I must admit that there are some relationship between them, to me they are very different in practical usage.

In this lesson, we will first introduce the projective plane, it is actually very simple. Next we will be going through two important notions: Cross Ratio and Perspectivity. Perspectivity can be thought of a geometric transformation and cross ratio is invariant under perspectivity. Meanwhile, perspectivity is unique determined by three of its values. The crux of these proofs are essentially trigonometry, which is why projective geometry can be thought of as "Trigonometry but Cleaner".

There are in general three ways to apply projective geometry in IMO. They are

1. Applying the classical "projective" theorems. They include Pascal's Theorem, Pappus' Theorem and Desargues Theorem. These theorems are impossible to prove using angle chasing. While they can be proved via Ceva' Menalaus, the proofs are very clumsy and complicated. Perspectivity and invariant of cross ratio give a very nice proof for these theorems. (However, in principle they are no different from Ceva/ Menalaus).

   Applying these theorems hence allow one to deduce collinearity/ concurrency which are otherwise difficult to obtain using common methods.

2. Using the theory of harmonic division, which are essentially just four points with a special cross ratio (which is why invariant of cross ratio is so useful). These quadruples of points satisfy some really nice Euclidean geometry properties.

3. Using the theory of pole/ polar, in particular La Hire's Theorem and Brocard's Theorem. At the end of the day they are the only useful ones. La Hire's theorem is useful in tangency problems, such as circumscribed quadrilaterals. Brocard's Theorem can be applied to any cyclic quadrilateral and is very related to the Miquel point configuration.

In fact, they correspond to Section 1-3 in Intermediate Notes. I have to dedicate this note for Ching Cow, since his notes are the place where I learnt this wonderful subject.

## 2 Set-up

### 2.1 Projective Plane

There are a couple of ways to define the projective plane, such as

- The set of lines in $\mathbb{R}^3$ through the origin.
- The 2-sphere, with antipodal points "identified"
- The upper hemisphere with the opposite points on the equator "identified".

These are easier to generalise to higher dimensions. They are equivalent (topologically) to the following, which is closer to applications in Euclidean Geometry. Think of it as the Euclidean plane with a "line at infinity". It can be thought of as a circle which "encloses" the whole plane. In particular, to each line $AB$ in the Euclidean plane we associate a "point of infinity" in the direction of line $AB$. Parallel lines are associated to the same point of infinity. The punchline is that

(i) There is a common point on every two lines

(ii) There is a common line passing through every two points.

Conics are important in projective geometry, however, for simplicity we will just assume all conics are circle/ lines.

### 2.2 Cross Ratio

Cross Ratio is defined for four points (which maybe ordinary points/ points at infinity) $A, B, C, D$ on a line or on a point. Using directed length, for four points on a line, the cross ratio is defined as

$$(A, B; C, D) = \frac{\frac{AC}{AD}}{\frac{BC}{BD}}$$

Similarly, if $A, B, C, D$ are four points on a circle, then

$$(A, B; C, D) = \frac{\frac{AC}{AD}}{\frac{BC}{BD}}$$

If you are familiar with trigonometry, the expressions $\frac{AC}{AD}$ and $\frac{BC}{BD}$ should be immediately recognisable, as the term appearing in the ratio lemma. In circles, they correspond to the sine of the corresponding angle at circumference.

### 2.3 Perspectivity

This is the main kind of geometric transformation.

**Definition 2.1** We will define it separably. Suppose $l_1, l_2$ are two lines and $P$ is point not on either of the lines, we can define a map from $l_1$ to $l_2$ which sends $X \in l_1$ to $PX \cap l_2$. (This is well-defined as any two lines has a unique intersection).

**Definition 2.2** Suppose $l$ is a line and $C$ is a circle, $P$ is a point on $C$. Define a map from $l$ to $C$ which sends each point $X$ on $C$ to $PX \cap l$. In particular it sends $P$ itself to $PP \cap l$, the intersection of the tangent at $P$ with $l$.

**Definition 2.3** Suppose $C_1, C_2$ are two circles, $P$ is a point on $C_1 \cap C_2$, define a map from $C_1$ to $C_2$ which sends each point $X$ on $C_1$ to $PX \cap C_2 \neq \{X\}$.

**Definition 2.4** We can also apply the principle of duality. Suppose $P, Q$ are two points and $AB$ is a line not containing $P$ and $Q$. Suppose $PC$ is a line, let $PC \cap AB = X$, then we map $PC$ to the line $QX$. This is called a perspectivity between pencil of lines.

**Definition 2.5** A sequence of perspectivity is called a projection.

We will write a perspectivity of lines as

$$l_1(A, B; C, D) \xrightarrow{P} l_2(E, F; G, H)$$

and a perspectivity of points as $P_1(P_1A, P_1B; P_1C, P_1D) \xrightarrow{Q_1Q_2} P_2(P_2A, P_2B; P_2C, P_2D)$

### 2.4 Main Result

**Theorem 2.1** The cross ratio is invariant under all perspectivities.
Proof.
Trigonometry. ■

**Theorem 2.2** Suppose $A_1, A_2, A_3$ are three points on a line $l_1$, and $B_1, B_2, B_3$ are three points on a line $l_2$. Then there exists a unique projection mapping $A_i$ to $B_i$
Proof.
Make two of the points equals first, then it is obvious. ■

**Theorem 2.3** A projective transformation between two lines is a perspectivity iff it fixes the intersection point. Dually, a projective transformation between two points is a perspectivity iff it fixes the line through the two points.

Despite being obvious, this allows us to prove theorems such as Desargues, Pascal and Pappus. It is not as useful (comparing to results in later sessions) in IMO though.

## 3 Projective Theorems

**Theorem 3.1** (Desargues Theorem) Suppose $\triangle ABC$ and $\triangle A_1B_1C_1$ are two triangles, then $X = AB \cap A_1B_1$, $Y = BC \cap B_1C_1$ and $Z = CA \cap C_1A_1$ are collinear if and only if $AA_1, BB_1, CC_1$ are concurrent.

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[Solution]
The only tools we have are Theorem 2.1, 2.2 and 2.3.

Suppose $A_1B_1, A_2B_2, A_3B_3$ are concurrent at a point $P$. Suppose $D = A_2A_3 \cap C_2C_3$, then

$$\begin{aligned} A_3(A_3C_1, A_3C_2; A_3C_3; A_3B_3) &\overset{A_1A_2}{\to} P(PB_2, PB_1, PC_3, PX) \\ &\overset{B_1B_2}{\to} B_3(B_2B_3, B_1B_3; B_3C_3, B_3A_3) \end{aligned}$$

By Theorem 2.3, this is a perspectivity. Hence $A_3C_1 \cap B_2B_3 = C_1$, $A_3C_2 \cap B_1B_3 = BC_2$ and $A_3C_3 \cap B_3C_3 = C_3$ are collinear.

Conversely, suppose $C_1, C_2, C_3$ are collinear, then

$$\begin{aligned} A_3A_2(A_3, A_2; A_1B_2 \cap A_2A_3, C_1) &\overset{A_1}{\to} C_2C_3(C_2, C_3; A_1B_1 \cap C_2C_3, C_1) \\ &\to (B_3, B_2; A_1B_1 \cap B_2B_3, C_1) \end{aligned}$$

Again by Theorem 1.2 this is a perspectivity, hence $A_2B_2 \cap ST \cap A_3B_3$ ∎.

Similarly we can prove the following two theorems

**Theorem 3.2** (Pascal's Theorem) Suppose $ABCDEF$ are six points on a circle, then $AB \cap DE$, $BC \cap EF$ and $FA \cap CD$ are collinear.

**Theorem 3.3** (Pappus' Theorem) Suppose $ACE$ and $BDF$ are two straight lines, then $AC \cap BD, AE \cap BF$ and $CE \cap DF$ are collinear.

**Remark.** When proving Theorem 3.1-3.3, Theorem 2.3 sounds like cheating and magically powerful. However, in practice they are rarely used (to the extent that I never knew their existence until last Tuesday while preparing). More often, we apply them indirectly through applying Theorem 3.1-3.3.

## 4 Harmonic Division

**Definition 4.1** Essentially they are quadruples of points with cross ratio equal to $-1$ (in directed length). That is,

$$\frac{\frac{CA}{CB}}{\frac{DA}{DB}} = -1$$

We say that $(A, B, C, D)$ is harmonic, or $A, B$ forms a harmonic conjugate pair with respect to $CD$.

What does this condition actually mean? This is related to the ratio lemma in trigonometry, recall that if $D$ is on the side $BC$ of $\triangle ABC$ then

$$\frac{BD}{CD} = \frac{\sin \angle BAD}{\sin \angle CAD} \cdot \frac{AB}{AC}$$

Hence it is natural to ask if the function $f : BC \to [0,\infty]$ given by $f(D) = \dfrac{BD}{CD}$ is injective. In fact, it is a two-to-one function, which maps harmonic conjugate pairs into the same value.

Amazingly, harmonic conjugate pairs are everywhere

**Theorem 4.1** The following are harmonic pairs.

(i) (Ceva-Menelaus Configuration) If $D, E, F$ are points on $BC, CA, AB$ of $\triangle ABC$ such that $AD, BE, CF$ are concurrent and $G = EF \cap BC$ then $(B, C; G, D) = -1$.

(ii) $(A, B; M_{AB}, \infty_{AB}) = -1$

(iii) In $\triangle PAB$, let $C, D$ be the foot of internal and external angle bisectors of $\angle CPD$, then $(A, B; C, D) = -1$

(iv) In $\triangle ABC$, suppose the $A$-symmedian intersect the circumcircle again at $D$, then $(A, D; B, C) = -1$

**Theorem 4.2** The following are equivalent for four collinear points $A, B, C, D$:

(i) $(A, B; C, D) = -1$

(ii) $AC \cdot AD = AB \cdot AM$

(iii) $MA \cdot MB = MC^2$.

**Example 4.1** Prove Theorem 4.1,4.2 and establish their converse.

**Remark.** Theorem 4.1,4.2 and their converses are incredibly useful. The general strategy is to obtain a group of harmonic points using 4.1/trigonometry, project them using perspectivity to obtain another group of harmonic points (note the invariance of cross ratio) and deduce geometric properties using the converse of 4.1/4.2.

**Remark.** Theorem 4.1(ii) are particularly useful for problems involving "Prove that $X$ is the midpoint of $YZ$" or "Prove that "$WX$ bisects $YZ$". In general there are very few other ways to deal with these relations other than appealing to Theorem 4.1 (ii)

**Remark.** Theorem 4.1 (iii) is kinda tricky as it is "extrinsic" in nature. Sometimes the point $P$ may not be so apparent and it takes some ingenuity to construct such a point.

**Remark.** Theorem 4.1(iv) is related to the symmedian configuration which you should be familiar with as level 2.

## 5 Pole and Polar

**Definition 5.1** Let $\Gamma$ be a circle, suppose $P$ is a point, take two lines $l_1, l_2$ through $P$ and suppose they intersect the circle again at $A_1, A_2$ and $B_1B_2$, by Pascal's Theorem, the line passing through $A_1B_2 \cap B_1A_2$ and $A_1B_1 \cap A_2B_22$ is independent of $P$, we call this line the polar of $P$.

**Definition 5.2** The pole of a line $l$ is the unique point whose polar is $l$.

**Theorem 5.1** Suppose $\Gamma$ is a circle with center $O$, and $P$ is a point, its polar is the line perpendicular to $OP$ and passes through $P'$, where $P'$ is the image of inversion of $P$ w.r.t. $\Gamma$.

**Definition 5.2** The pole of a line $l$ is the unique point whose polar is $l$.

**Theorem 5.2** (Brocard's Theorem) Suppose $ABCD$ is a cyclic quadrilateral, then let $AB \cap CD = E, AC \cap BD = F$ and $AD \cap BC = G$. Let $O$ be the center of $(ABCD)$, then among $E, F, G$, the polar of each point is the line form by the other two. In particular, $OE \perp FG, OF \perp EG$ and $OG \perp EF$.

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[Solution]
Obvious.

**Theorem 5.3** (La Hire's Theorem) $A$ lies on the polar of $B$ iff $B$ lies on the polar of $A$

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[Solution]
Obvious.

**Remark.** Theorem 5.3 is useful for problems involving multiple tangent lines, such as circumscribed quadrilateral (see IMO shortlist compilations).

**Remark.** Theorem 5.2 is useful for more general configurations, mostly those for complete quadrilaterals.

## 6 Further Remarks

There are three more advanced topics regarding projective geometry. Unfortunately, I failed to understand them back when I was a student. So you may have to consult other references:

1. Moving Points
2. Desargues Involution Theorem/ Dual of Desargues Involution Theorem
3. Projective Transformation

Moving points seem quite popular these days, DIT/DDIT can be found on Chapter 4 of Intermediate Notes, and projective transformation can be found on Chapter 9.7 of EGMO.

## 7 Exercises

1. (CHKMO 2011 Problem 2) The incircle of a triangle $ABC$ touches three sides $BC, CA$ and $AB$ at $D, E$ and $F$ respectively. The line segments $BE$ and $CF$ intersect the incircle at $Y$ and $Z$ respectively. SUppose the lines $FE$ and $BC$ intersect externally at $X$. Prove that the three points $X, Y$ and $Z$ are collinear.

2. (CWMO 2005 Problem 2) Two tangents $PA, PB$ are drawn from a point $P$ outside a circle where $A, B$ are contact points. Another secant to the circle through $P$ intersects the circle at the points $C, D$. A line through $B$ parallel to $PA$ intersects the lines $AC, AD$ at $E, F$ respectively. Prove that $BE = BF$.

3. (CSMO 2007 Problem 2) Let $C, D$ be two arbitrary points on a semicircle with centre $O$ and diameter $AB$. The tangent to circle $O$ at the point $B$ intersects the line $CD$ at $P$. Suppose the line $PO$ meets the lines $CA, AD$ at the points $E, F$ respectively. Prove that $OE = OF$.

4. (CMO 1996 Problem 1) Let $H$ be the orthocentre of an acute $\triangle ABC$. Construct the tangents $AP, AQ$ from $A$ to the circle with $BC$ as a diameter, where $P, Q$ are the contact points. Prove that $P, H, Q$ are collinear.

# 8 Problems

1. (Korea MO 2012 P4) Let $ABC$ be an acute triangle. Let $H$ be the foot of perpendicular from $A$ to $BC$. $D, E$ are the points on $AB, AC$ and let $F, G$ be the foot of perpendicular from $D, E$ to $BC$. Assume that $DG \cap EF$ is on $AH$. Let $P$ be the foot of perpendicular from $E$ to $DH$. Prove that $\angle APE = \angle CPE$.

2. (APMO 2012 Problem 4) Let $ABC$ be an acute triangle. Denote by $D$ the foot of the perpendicular line drawn from the point $A$ to the side $BC$, by $M$ the midpoint of $BC$, and by $H$ the orthocenter of $ABC$. Let $E$ be the point of intersection of the circumcircle $\Gamma$ of the triangle $ABC$ and the half line $MH$, and $F$ be the point of intersection (other than $E$) of the line $ED$ and the circle $\Gamma$. Prove that $\frac{BF}{CF} = \frac{AB}{AC}$ must hold.

3. (APMO 2013 Problem 5) Let $ABCD$ be a quadrilateral inscribed in a circle $\omega$, and let $P$ be a point on the extension of $AC$ such that $PB$ and $PD$ are tangent to $\omega$. The tangent at $C$ intersects $PD$ at $Q$ and the line $AD$ at $R$. Let $E$ be the second point of intersection between $AQ$ and $\omega$. Prove that $B$, $E$, $R$ are collinear.

4. (APMO 2021 Problem 3) Let $ABCD$ be a cyclic convex quadrilateral and $\Gamma$ be its circumcircle. Let $E$ be the intersection of the diagonals of $AC$ and $BD$. Let $L$ be the center of the circle tangent to sides $AB$, $BC$, and $CD$, and let $M$ be the midpoint of the arc $BC$ of $\Gamma$ not containing $A$ and $D$. Prove that the excenter of triangle $BCE$ opposite $E$ lies on the line $LM$.

5. (CHKMO 2021 Problem 2) In acute $\triangle ABC, BC > AB$. Let $D$ be the point on the side $AC$ such that $BD = BA$. Let $M$ be the midpoint of $BC$. The tangents at $D$ and $M$ to the circumcircle of $\triangle CDM$ intersect at point $E$. The line $BE$ intersects the side $AC$ at $F$. Prove that $A, B, M, F$ are concyclic.

6. (ELMO shortlist 2019 G4) Let triangle $ABC$ have altitudes $BE$ and $CF$ which meet at $H$. The reflection of $A$ over $BC$ is $A'$. Let $(ABC)$ meet $(AA'E)$ at $P$ and $(AA'F)$ at $Q$. Let $BC$ meet $PQ$ at $R$. Prove that $EF \parallel HR$.

7. (Sharygin 2021 Q14) Let $\gamma_A, \gamma_B, \gamma_C$ be excircles of triangle $ABC$, touching the sides $BC$, $CA$, $AB$ respectively. Let $l_A$ denote the common external tangent to $\gamma_B$ and $\gamma_C$

distinct from $BC$. Define $l_B, l_C$ similarly. The tangent from a point $P$ of $l_A$ to $\gamma_B$ distinct from $l_A$ meets $l_C$ at point $X$. Similarly the tangent from $P$ to $\gamma_C$ meets $l_B$ at $Y$. Prove that $XY$ touches $\gamma_A$.

8. (All-Russian MO 2017 Grade 10 Problem 8) In a non-isosceles triangle $ABC$,$O$ and $I$ are circumcenter and incenter,respectively.$B'$ is reflection of $B$ with respect to $OI$ and lies inside the angle $ABI$.Prove that the tangents to circumcirle of $\triangle BB'I$ at $B'$,$I$ intersect on $AC$.

9. (All-Russian MO 2016 Grade 10 Problem 8) In acute triangle $ABC$,$AC < BC$,$M$ is midpoint of $AB$ and $\Omega$ is it's circumcircle.Let $C'$ be antipode of $C$ in $\Omega$. $AC'$ and $BC'$ intersect with $CM$ at $K, L$,respectively.The perpendicular drawn from $K$ to $AC'$ and perpendicular drawn from $L$ to $BC'$ intersect with $AB$ and each other and form a triangle $\Delta$.Prove that circumcircles of $\Delta$ and $\Omega$ are tangent.

10. (Korea MO 2019 P2) For a rectangle $ABCD$ which is not a square, there is $O$ such that $O$ is on the perpendicular bisector of $BD$ and $O$ is in the interior of $\triangle BCD$. Denote by $E$ and $F$ the second intersections of the circle centered at $O$ passing through $B, D$ and $AB, AD$. $BF$ and $DE$ meets at $G$, and $X, Y, Z$ are the foots of the perpendiculars from $G$ to $AB, BD, DA$. $L, M, N$ are the foots of the perpendiculars from $O$ to $CD, BD, BC$. $XY$ and $ML$ meets at $P$, $YZ$ and $MN$ meets at $Q$. Prove that $BP$ and $DQ$ are parallel.

11. (Geometry in Figures 4.1.33) In $\triangle ABC$, $D$ is the foot of altitude from $A$ and $E$ is a point on $AD$. Let $F = AE \cap AB$ and $G = AE \cap AC$. Suppose $R$ is a point on $BC$ and $J, I$ are the foot of altitudes from $D$ to $FR, GR$ respectively. Show that $J, I, E, A$ are concyclic.

12. (IMO shortlist 2012 G7) Let $ABCD$ be a convex quadrilateral with non-parallel sides $BC$ and $AD$. Assume that there is a point $E$ on the side $BC$ such that the quadrilaterals $ABED$ and $AECD$ are circumscribed. Prove that there is a point $F$ on the side $AD$ such that the quadrilaterals $ABCF$ and $BCDF$ are circumscribed if and only if $AB$ is parallel to $CD$.